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Course Name – Data Analytics With GenAI

Subject – PowerBI

Topic – Advanced Visuals in PowerBI

PowerBI File : [File Link](#)

Q1. What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

Answer 1.

In Power BI, we have different ways to filter data depending on how we want to use it.

Basic Filter is the simplest one. It's available in the Filters pane and works by letting you check/uncheck specific values. For example, if I want to show only 2023 data, I just tick "2023" from the list. It's straightforward but not very flexible.

Advanced Filter is also in the Filters pane but gives more control. Instead of just selecting values, you can write conditions like "Sales is greater than 50000" or "Product name contains Lap". You can also combine two conditions using AND/OR. It's useful when you need logic-based filtering.

Slicer is different from both, it's an actual visual that sits on the report canvas. Users can see it and interact with it directly. For example, a dropdown or list showing all regions, and clicking "North" filters the whole page. It's the most user-friendly option because non-technical users can also use it without going into the Filters pane.

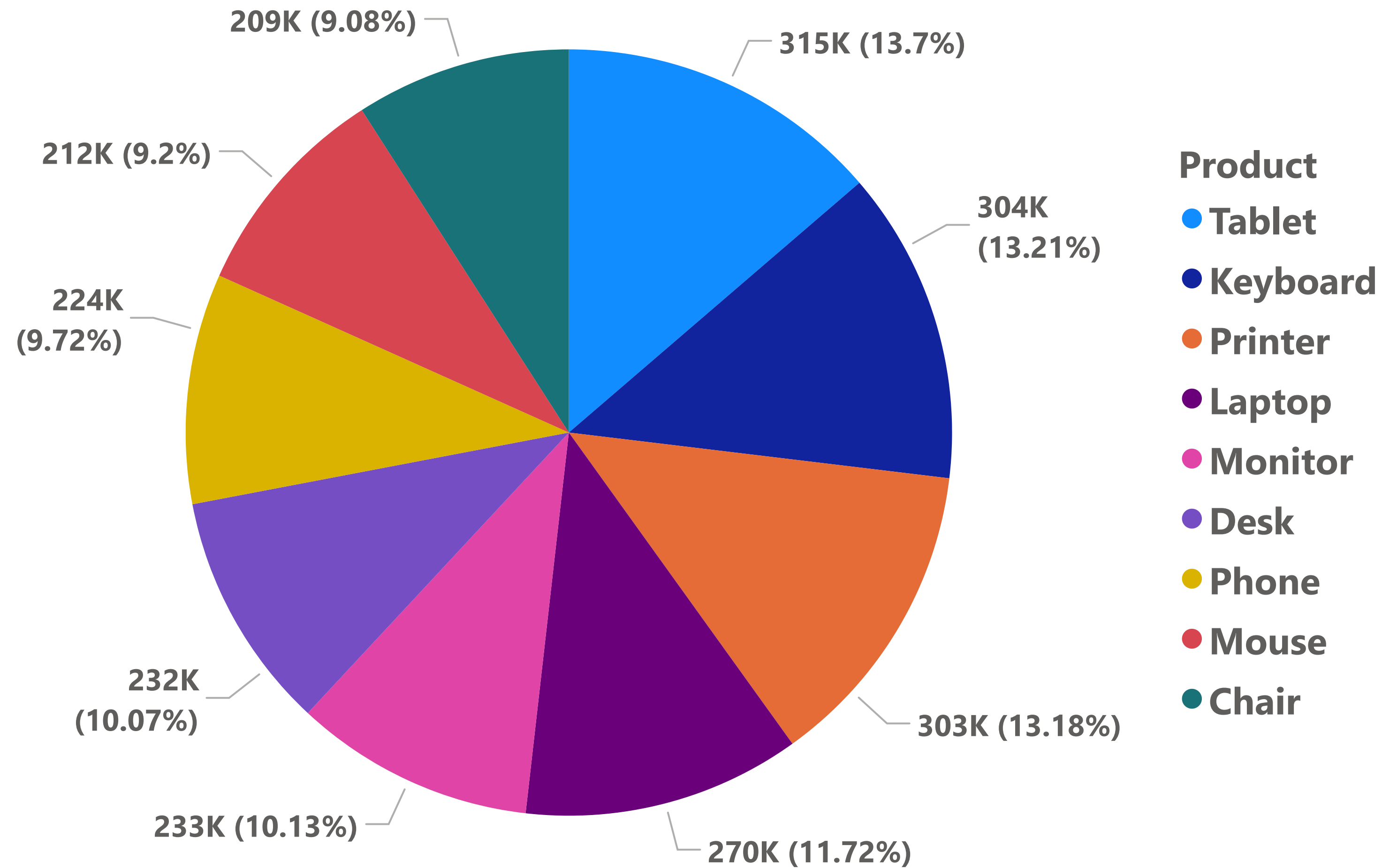
So basically, Basic Filter is quick and simple, Advanced Filter is for conditional logic, and Slicer is for end-user interactivity on the dashboard itself.

Q2. Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.

Region	Sum of Sales
East	71681
North	118922
South	264205
West	272091
Total	726899

Q3. Create a Pie Chart of Sales by Product and add Slicer that allows users to filter the dataset by Region.

Sales by Product



Q4. What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

Answer 4

So basically in Power BI, filters work at three different levels depending on how widely you want them to apply.

Visual-Level Filter only affects the one visual you apply it to. Like if I have 3 charts on a page and I apply a filter on one bar chart to show only Electronics, the other two charts are completely unaffected. It's useful when you want one specific visual to show different data than the rest.

Page-Level Filter applies to everything on that particular page. So if I set Country = India on page 1, all visuals on page 1 will show only India data, but page 2 and page 3 are not affected at all.

Report-Level Filter is the broadest one. Whatever filter you set here applies to every single visual on every single page of the report. For example in our Sales dataset, if I set Country = India as a report-level filter, even if I go to page 3, it will still show only India records.

A simple way I remember it — Visual is the smallest scope, Page is medium, and Report is the widest. You choose based on how broadly you want the filter to work.

Q5. Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India.

Country	Sum of Sales
India	2300843

Q6. What is the role of Drill-Down in Power BI? How does it improve data exploration?

Answer 6

Drill-Down allows users to navigate from a high-level summary to increasingly granular details within the same visual, without needing separate pages or visuals.

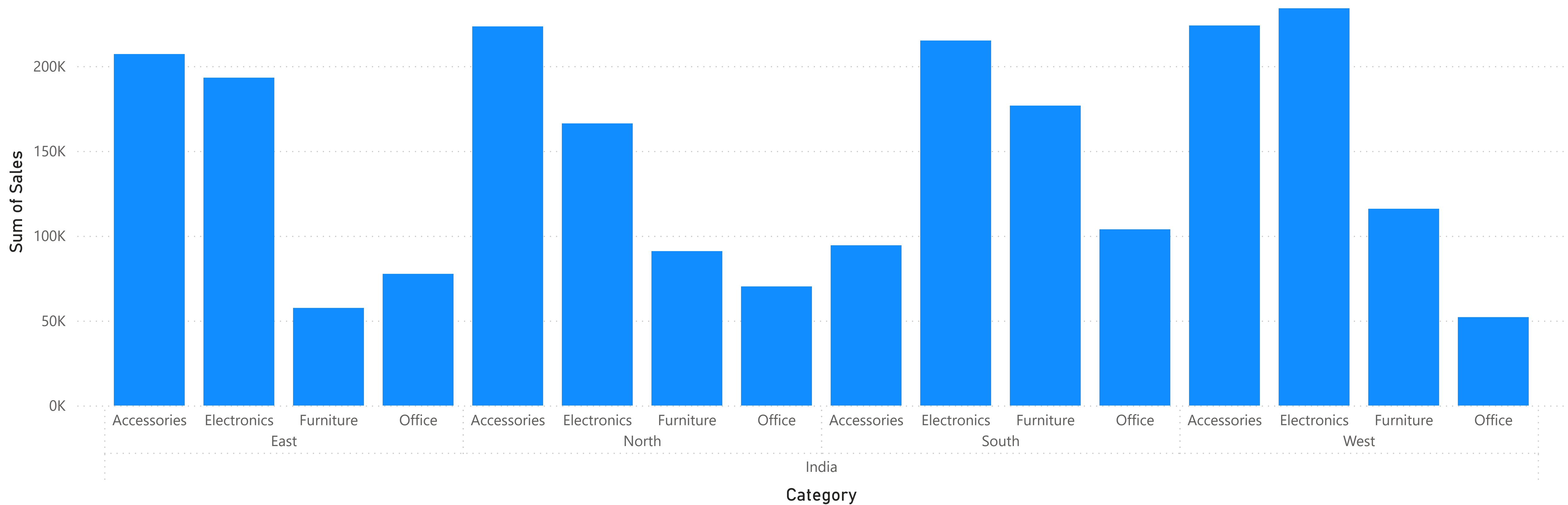
How it improves exploration:

- . Users start at an aggregate level (e.g., Country) and click to go deeper (Region → Category).
- . One visual handles multiple levels of data granularity, reducing clutter.
- . Business users can explore on their own without technical help.
- . Filters and context are preserved as you drill deeper.

Example: A bar chart shows total sales by Category. Clicking "Electronics" drills into Laptop, Phone, Tablet, etc.

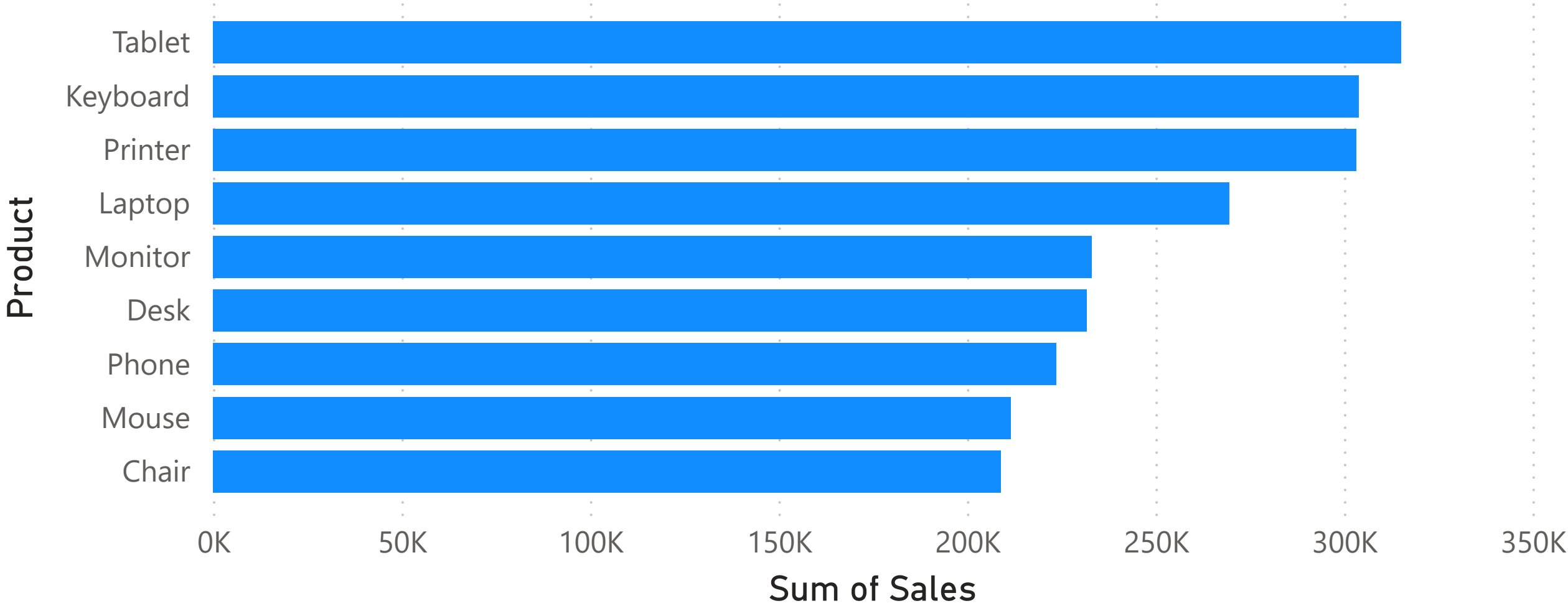
Q7. Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales at different levels.

Sum of Sales by Country, Region and Category

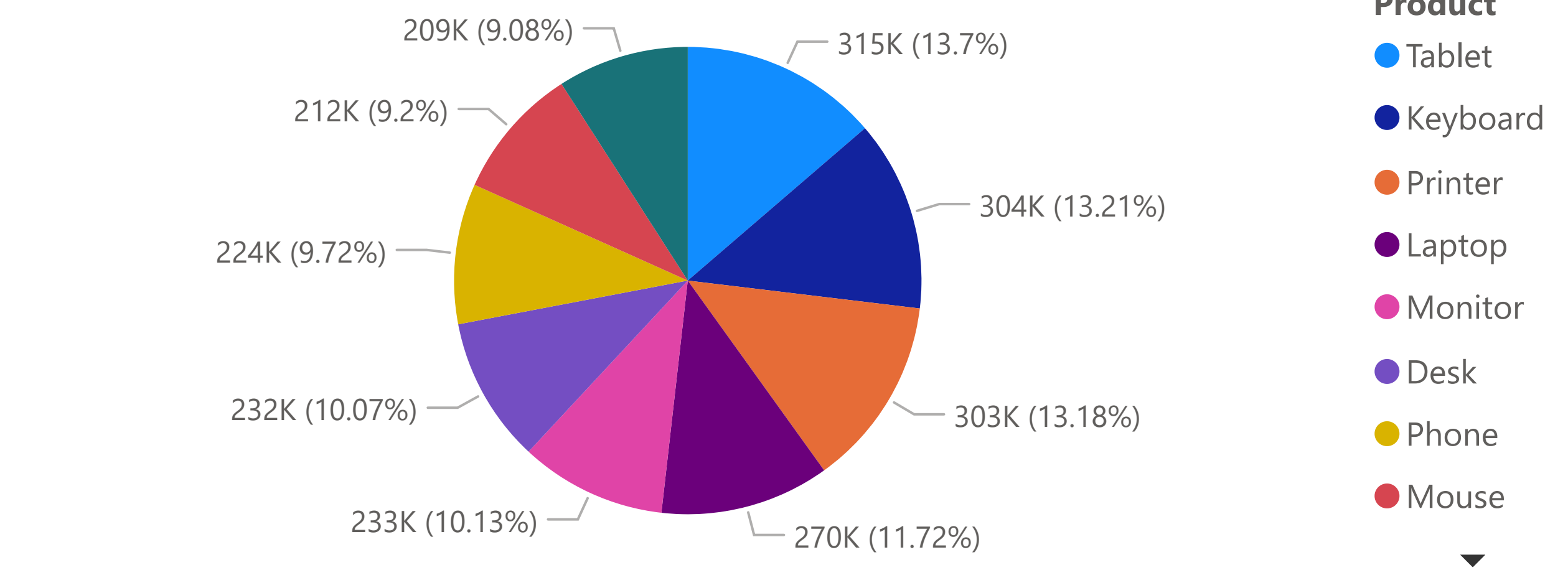


Q8. Implement Drill-Through so that selecting a Product navigates to a detailed report page showing regionwise sales.

Sum of Sales by Product



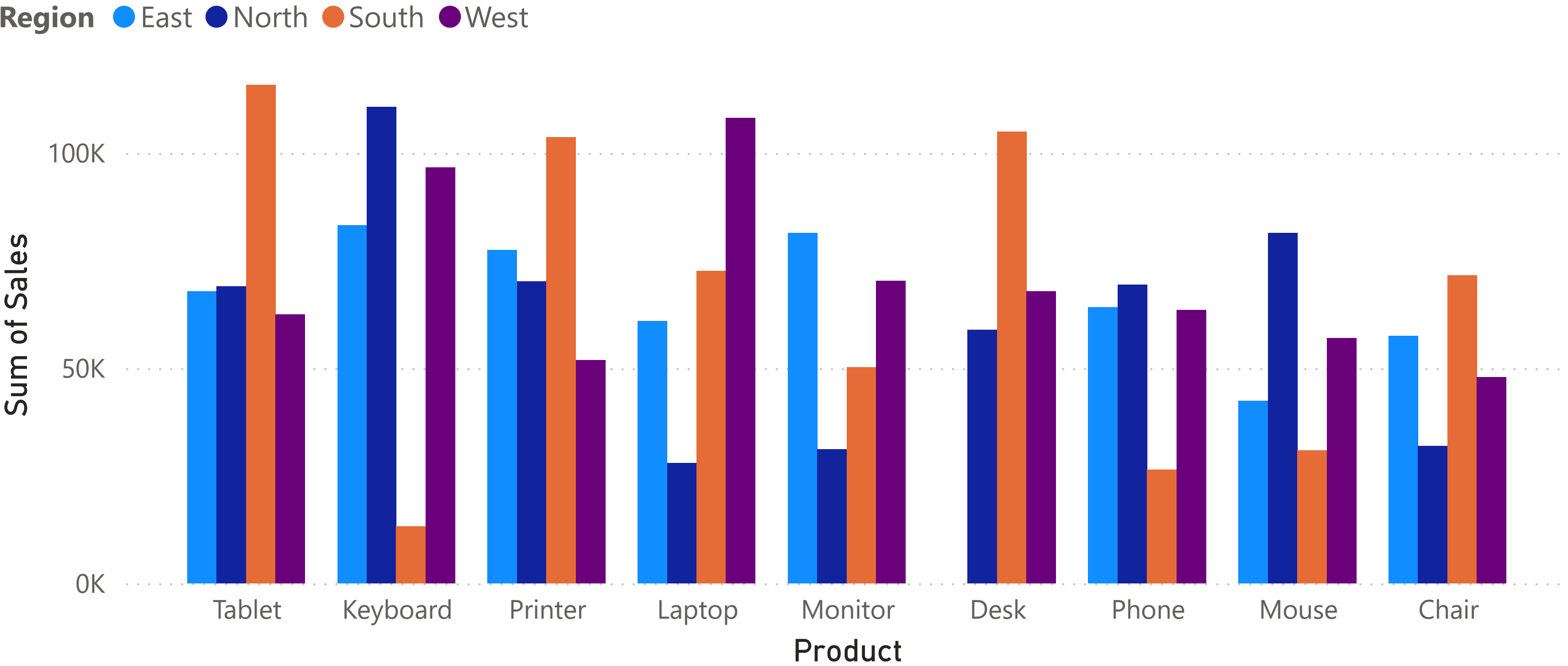
Sum of Sales by Product



Product Category Sum of Quantity Sum of Sales

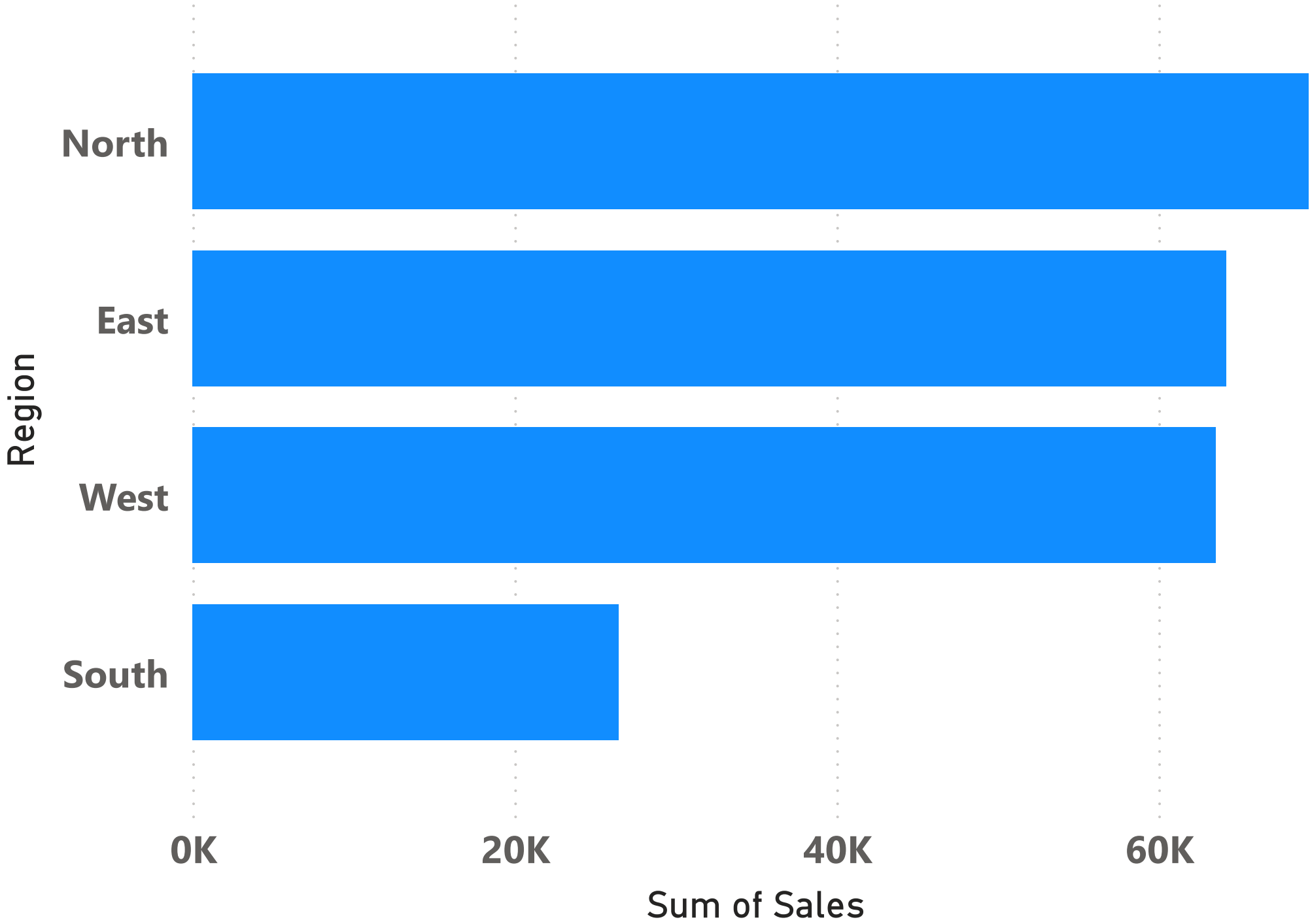
Chair	Furniture	104	208996
Desk	Furniture	105	231793
Keyboard	Accessories	145	303877
Laptop	Electronics	160	269552
Monitor	Accessories	114	233033
Mouse	Accessories	116	211644
Phone	Electronics	103	223560
Printer	Office	151	303220
Tablet	Electronics	132	315168
Total		1130	2300843

Sum of Sales by Product and Region

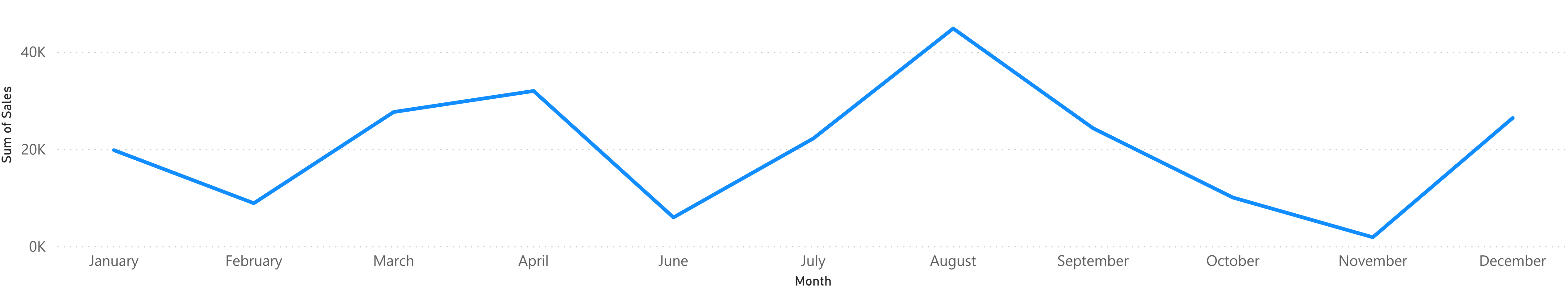




Sum of Sales by Region



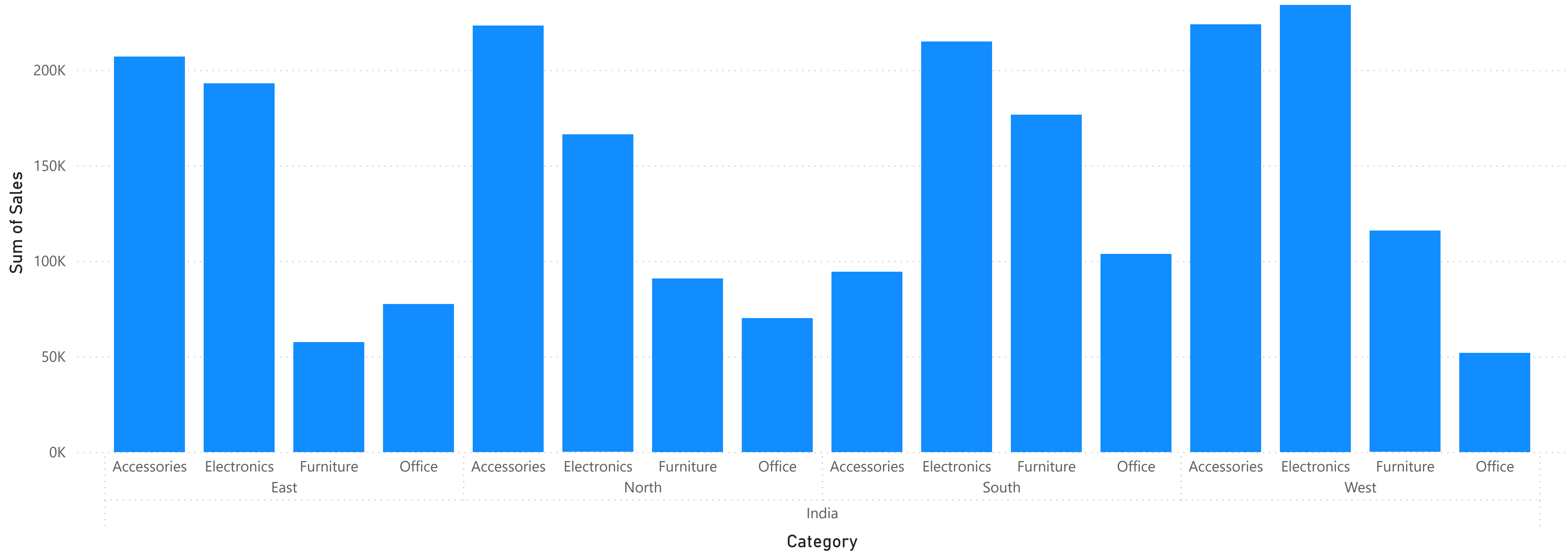
Sum of Sales by Month



Order_ID	Year	Month	Customer_Name	Product	Region	Sum of Sales
1002	2023	February	Bhavya	Phone	South	8000
1146	2022	January	Customer_126	Phone	South	19755
1160	2024	March	Customer_140	Phone	West	10709
1176	2024	March	Customer_156	Phone	East	7382
1181	2023	September	Customer_161	Phone	West	8443
1196	2022	November	Customer_176	Phone	North	1842
1203	2022	October	Customer_183	Phone	West	10010
1206	2024	March	Customer_186	Phone	West	4187
1216	2024	September	Customer_196	Phone	East	15828
1225	2023	April	Customer_205	Phone	West	4100
1228	2024	December	Customer_208	Phone	North	17850
Total						223560

Q9. Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy.

Sum of Sales by Country, Region and Category



Q10. Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example.

Answer 10

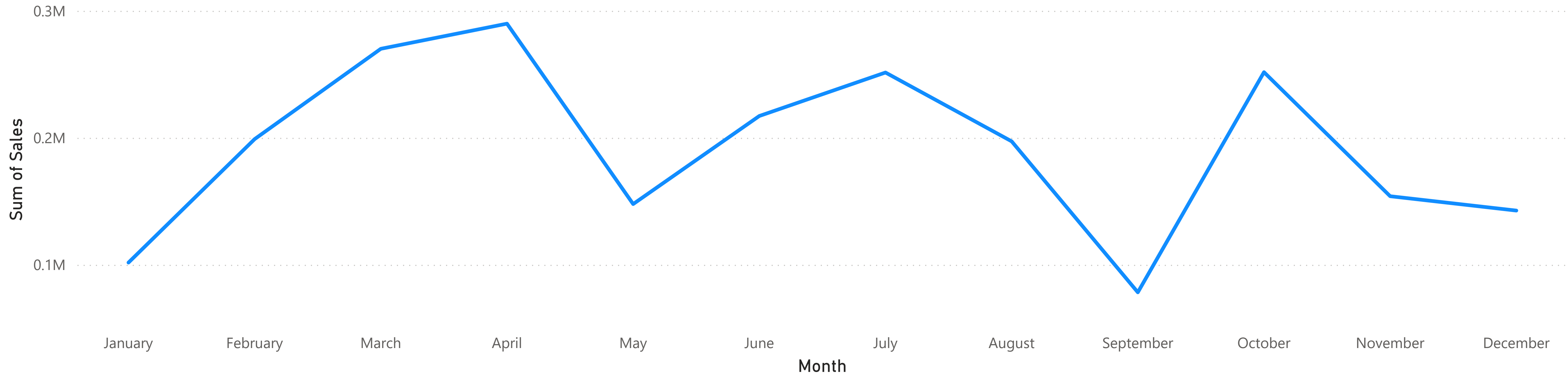
Drill-Down stays on the same page and same visual. You go deeper into the data. Like in our Sales dataset — I have a bar chart showing sales by Year. I click on 2023, it shows me Q1, Q2, Q3, Q4. I click Q3, it shows July, August, September. I never left the chart, just went deeper.

Drill-Through takes you to a completely different page. Like I right-click on "Laptop" in my product chart and it takes me to the Product Detail page showing only Laptop data. I moved from one page to another.

Simple real world example: Think of it like Google Maps. Drill-Down is like zooming in on the map — you're still on the same map, just seeing more detail. Drill-Through is like clicking on a restaurant and it opens a completely new page with reviews, menu, timings — different page, but related to what you clicked.

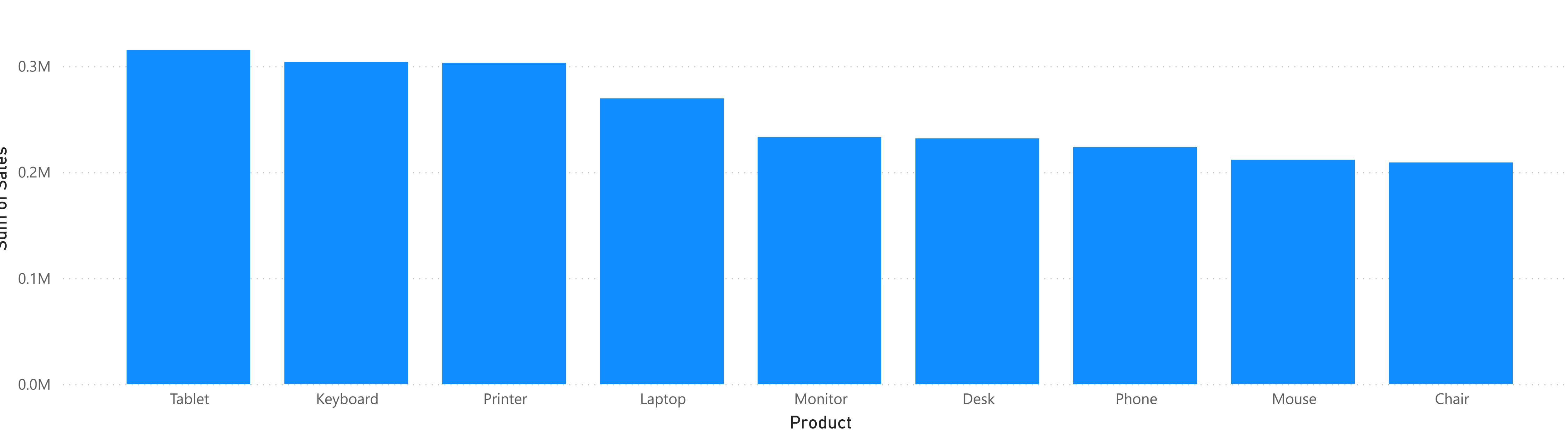
Q11. Use the Drill-Down feature to analyze sales trends from Year → Quarter → Month → Day.

Sum of Sales by Month



Q12. Apply a Measure-Based Filter to display only those products where total sales exceed ₹50,000.
Hint: Create a Total Sales measure and apply the filter using the measure, not the Sales column.

Sum of Sales by Product



Q13 What is the “Keep All Filters” option in Drill-Through? How does it affect user interaction

Answer 13

When I drill-through from main page to Product Detail page, there are two scenarios:

Keep All Filters = OFF (default) Suppose on main page I have selected Region = North in slicer and then I drill through on Laptop. The Product Detail page will show Laptop data from ALL regions, not just North. Only the product filter carries over, everything else is ignored.

Keep All Filters = ON Same situation — I have Region = North selected and drill through on Laptop. Now the Product Detail page shows Laptop sales from North region only. All my slicer selections and filters travel with me to the new page.

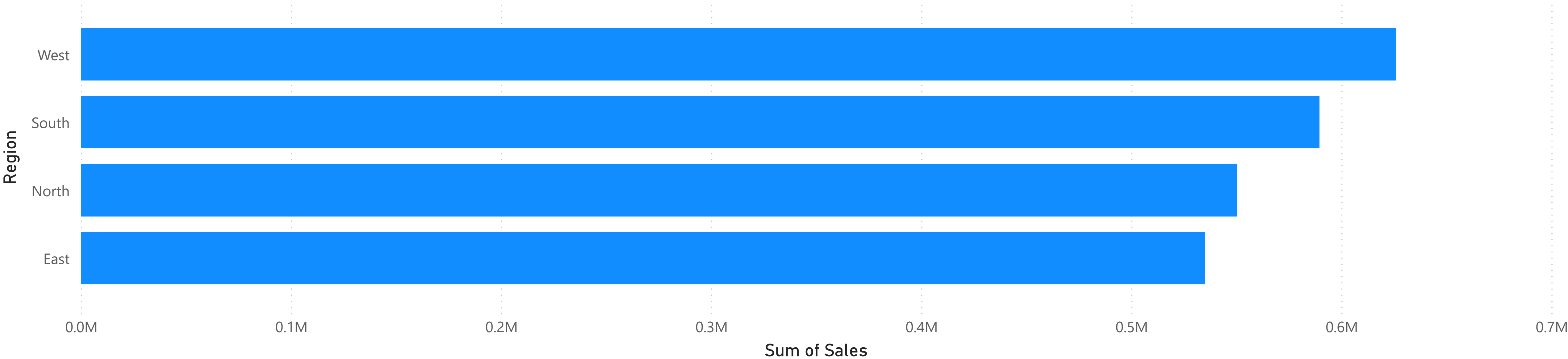
How to turn it on:

- Go to your drill-through destination page (Product Detail)
- In Visualizations pane, under Drill through section
- You'll see a toggle that says "**Keep all filters**"
- Turn it ON

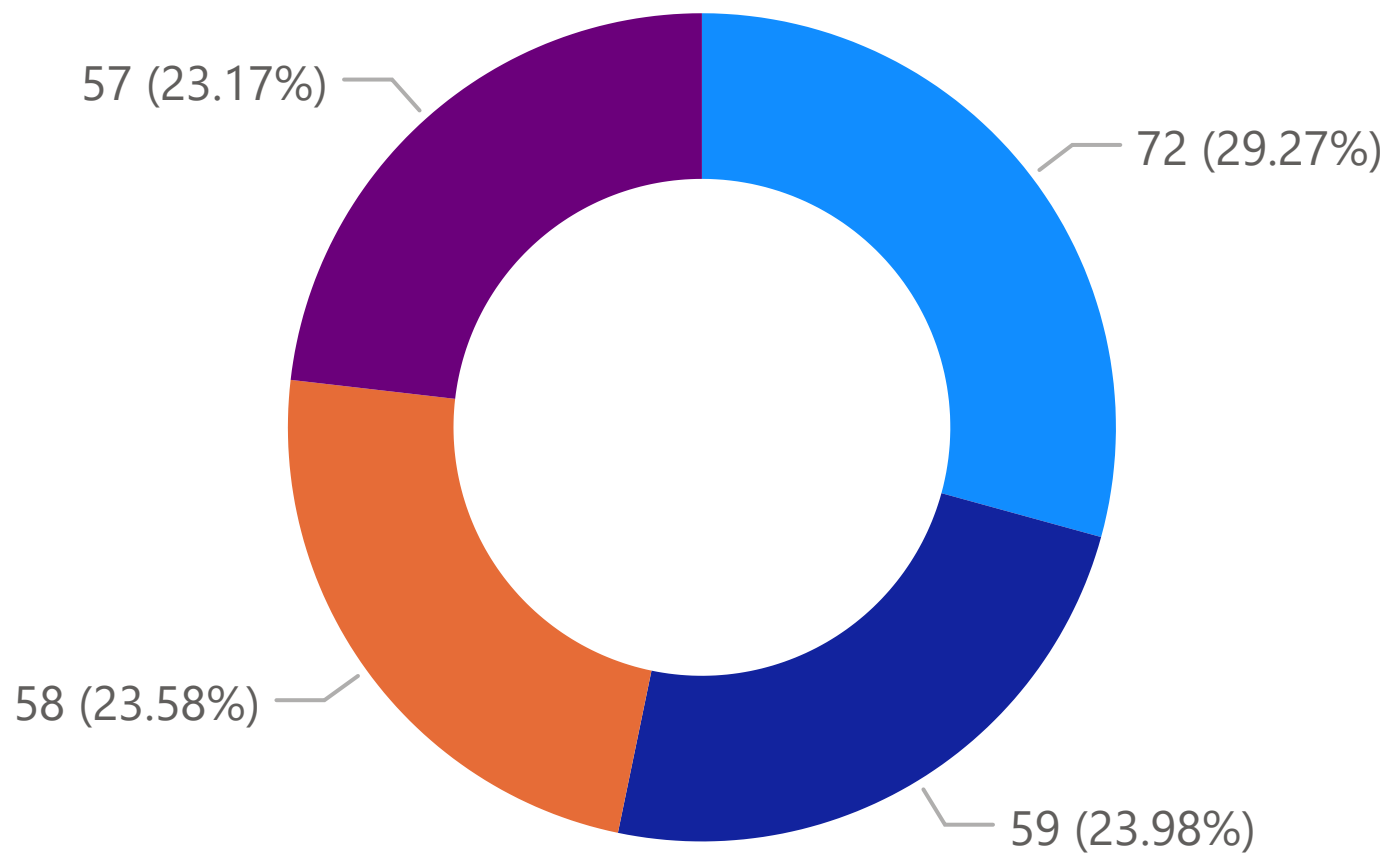
Which one to use: If you want users to see the complete picture of a product regardless of what they had selected, keep it OFF. If you want the context to be preserved, turn it ON. In our Sales dataset, turning it ON makes more sense because if someone was looking at North region and drills into Laptop, they probably want to see North region Laptop data specifically.

Q14 Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends over time.

Sum of Sales by Region



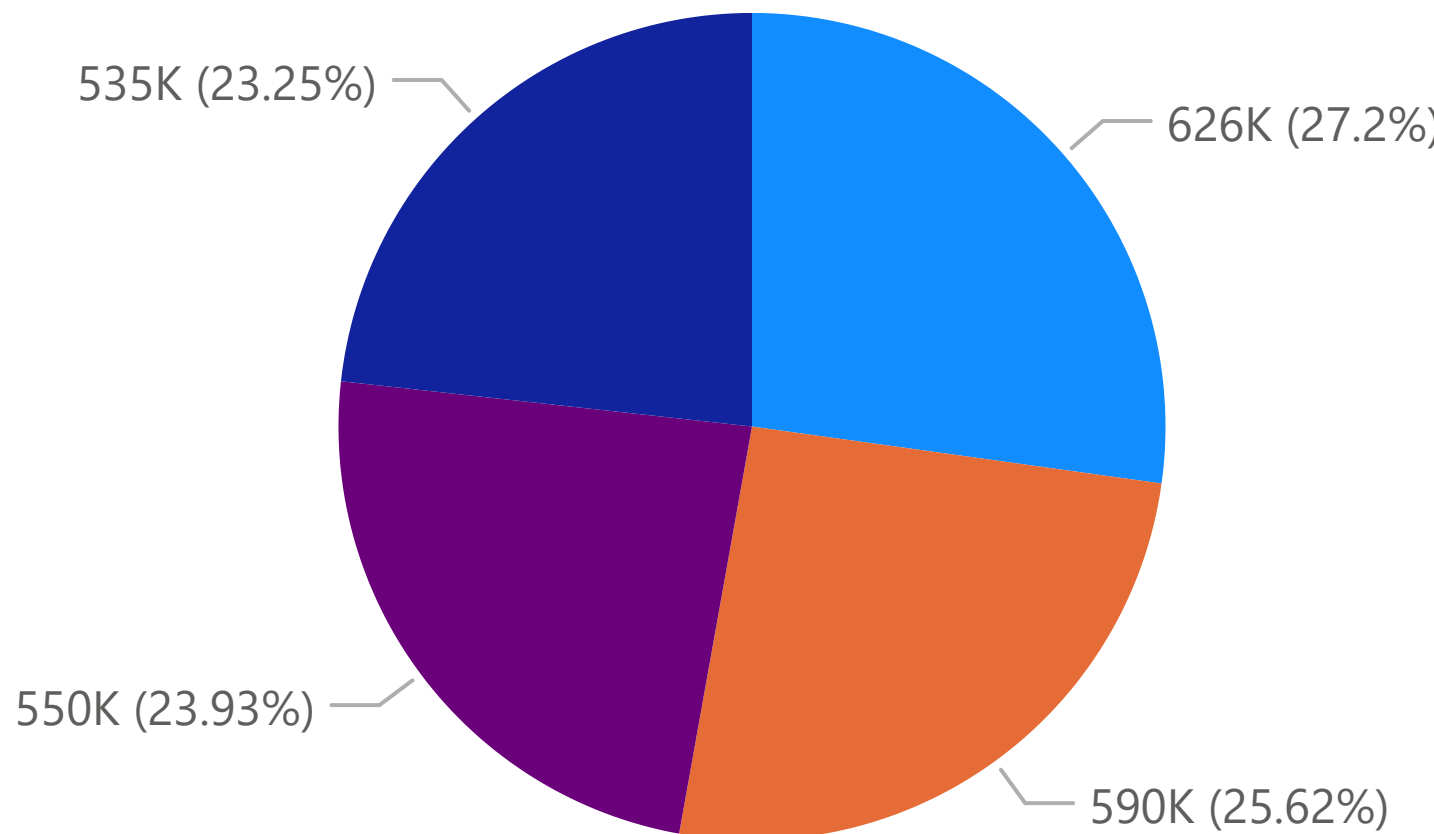
Count of Order_ID by Region



Region

- West
- East
- South
- North

Sum of Sales by Region



Region

- West
- South
- North
- East



535K

Total Sales

59

Count of Order ID

9.07K

Average of Sales

The chart displays the monthly sum of sales over a three-year period. The Y-axis, labeled 'Sum of Sales', ranges from 0K to 60K in increments of 20K. The X-axis, labeled 'Year', shows dates from July 2021 to July 2024. The sales data is represented by a blue line. The chart shows a steady increase in sales from July 2021 to early 2022, followed by a period of fluctuation with peaks around 38K in mid-2022 and 40K in early 2023. A significant peak of approximately 52K occurred in early 2024, followed by a sharp decline and a subsequent recovery to around 42K by July 2024.

Date	Sum of Sales (K)
Jul 2021	0
Jan 2022	5
Jul 2022	38
Jan 2023	10
Jul 2023	1
Jan 2024	52
Jul 2024	42

Product	Sum of Sales (K)
Keyboard	84
Monitor	82
Printer	78
Tablet	69
Phone	65
Laptop	61
Chair	58
Mouse	43

Order_ID	Year	Quarter	Month	Day	Customer_Name	Product	Sum of Quantity	Sum of Sales
1003	2023	Qtr 1	March	20	Charan	Tablet	1	500
1007	2022	Qtr 3	July	15	Garima	Phone	2	850
1011	2023	Qtr 2	June	12	Kavya	Laptop	1	1250
1019	2021	Qtr 1	February	25	Tanmay	Monitor	3	450
1021	2024	Qtr 2	May	10	Customer_1	Printer	8	5890
1022	2022	Qtr 2	May	2	Customer_2	Keyboard	8	17350
1023	2024	Qtr 2	May	21	Customer_3	Keyboard	6	16523
1027	2024	Qtr 2	April	11	Customer_7	Printer	5	18412
1035	2022	Qtr 4	October	23	Customer_15	Chair	5	9029
1038	2023	Qtr 3	August	16	Customer_18	Tablet	3	891
1043	2024	Qtr 3	August	18	Customer_23	Tablet	3	15448
Total							275	535012

2M

Total Sales

246

Total Orders

1K

Total Quantity

Year

- ☐ 2021
- ☐ 2022
- ☐ 2023
- ☐ 2024

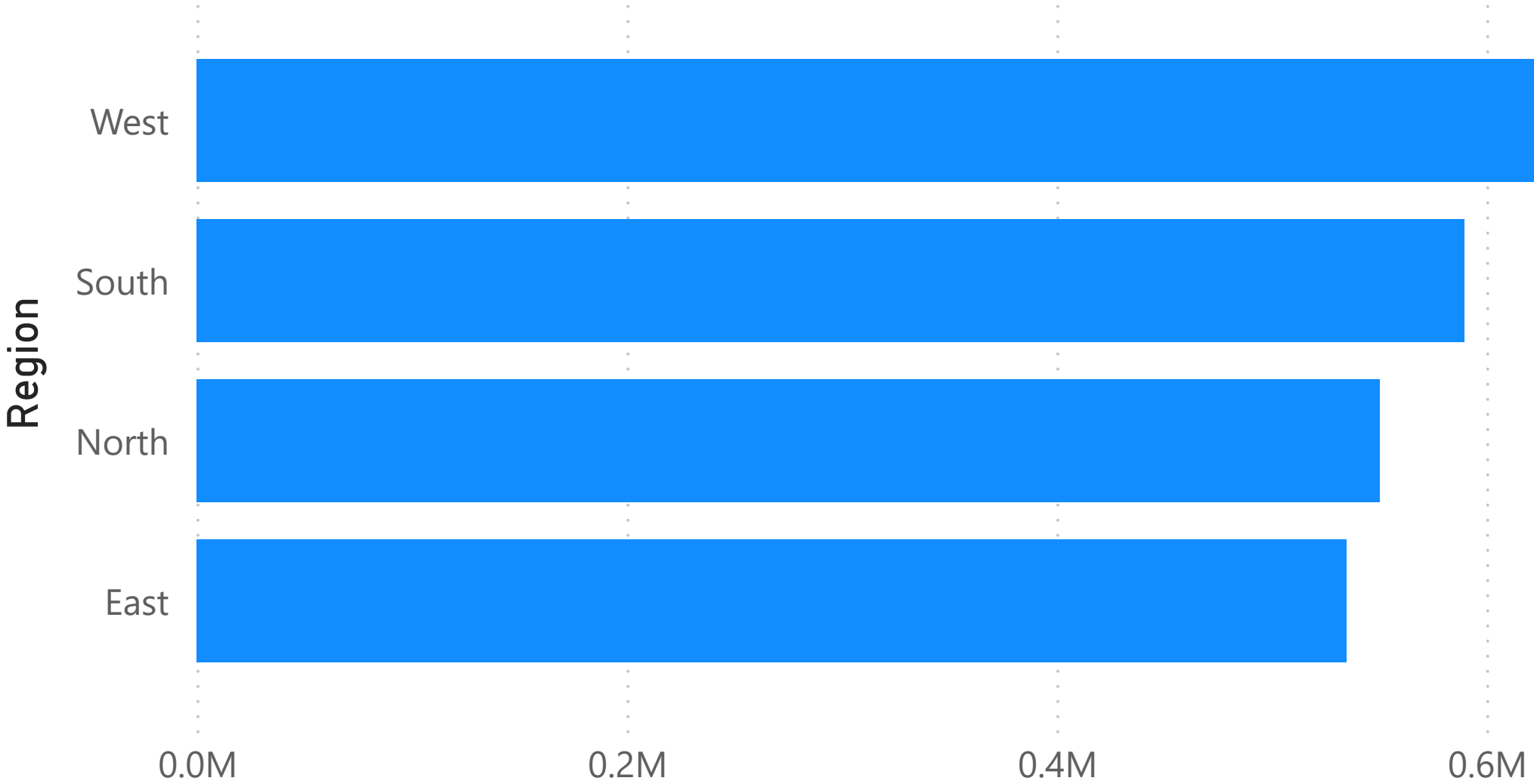
Category

- ☐ Accessories
- ☐ Electronics
- ☐ Furniture
- ☐ Office

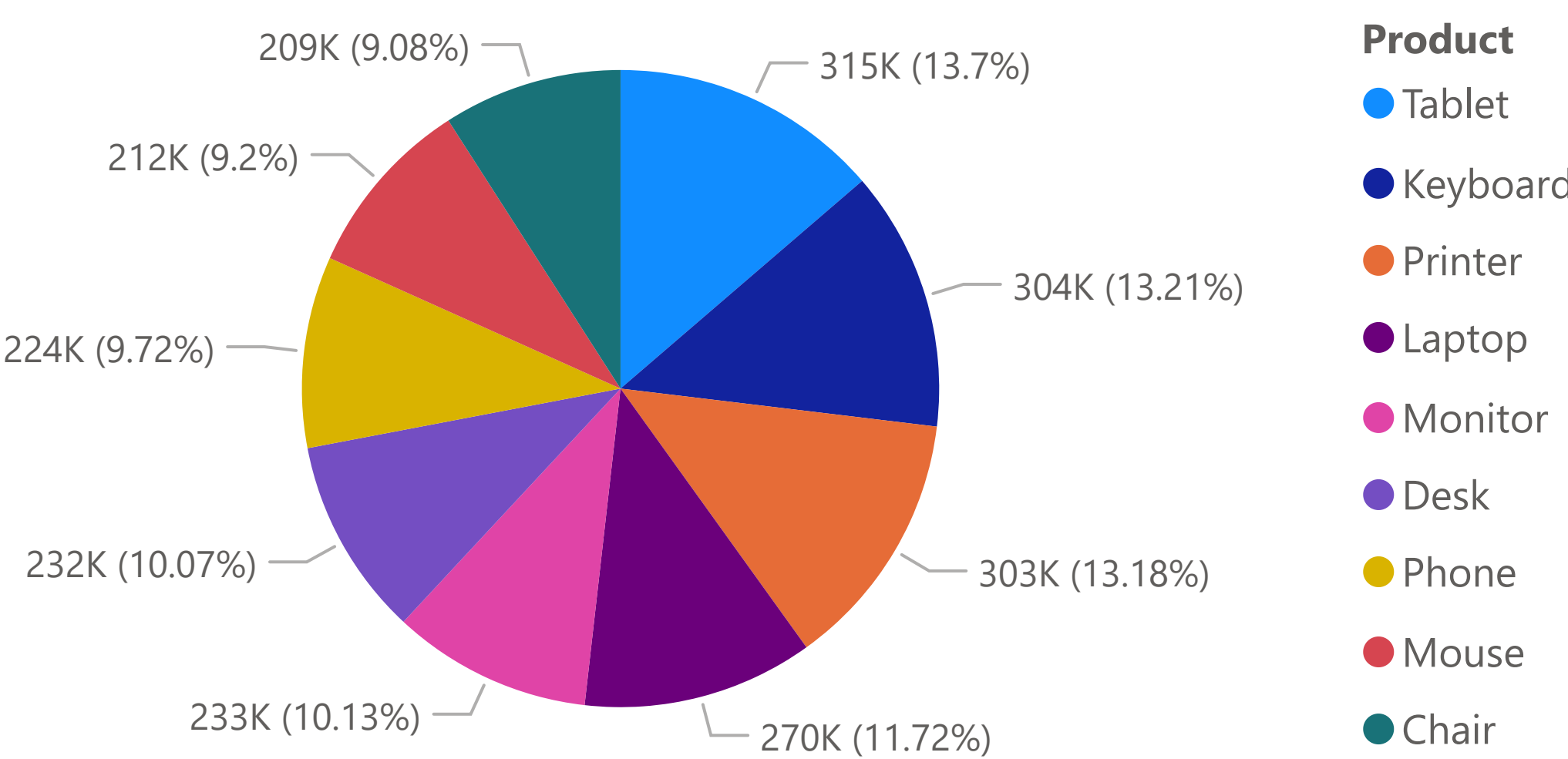
Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

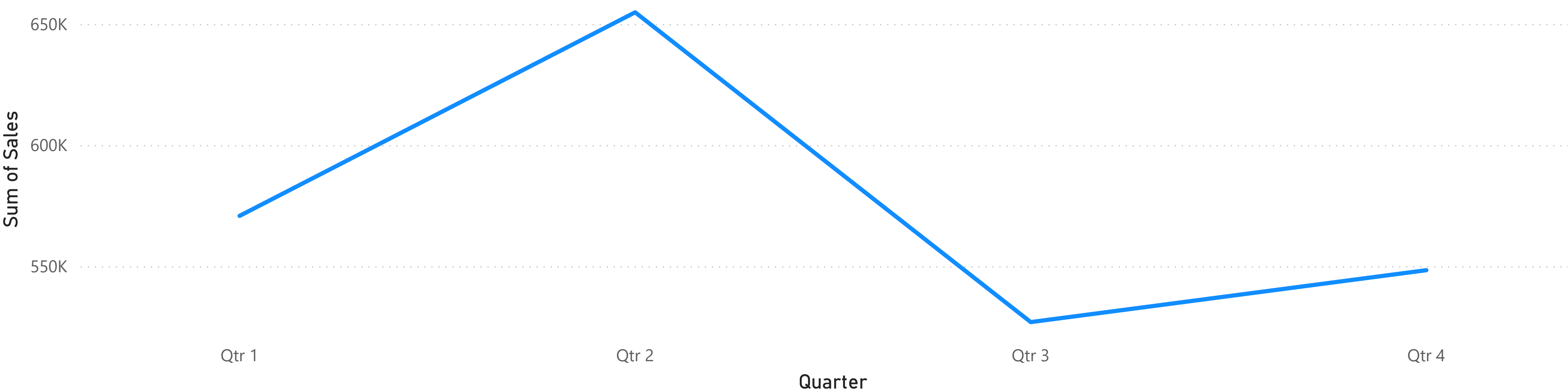
Sum of Sales by Region



Sum of Sales by Product



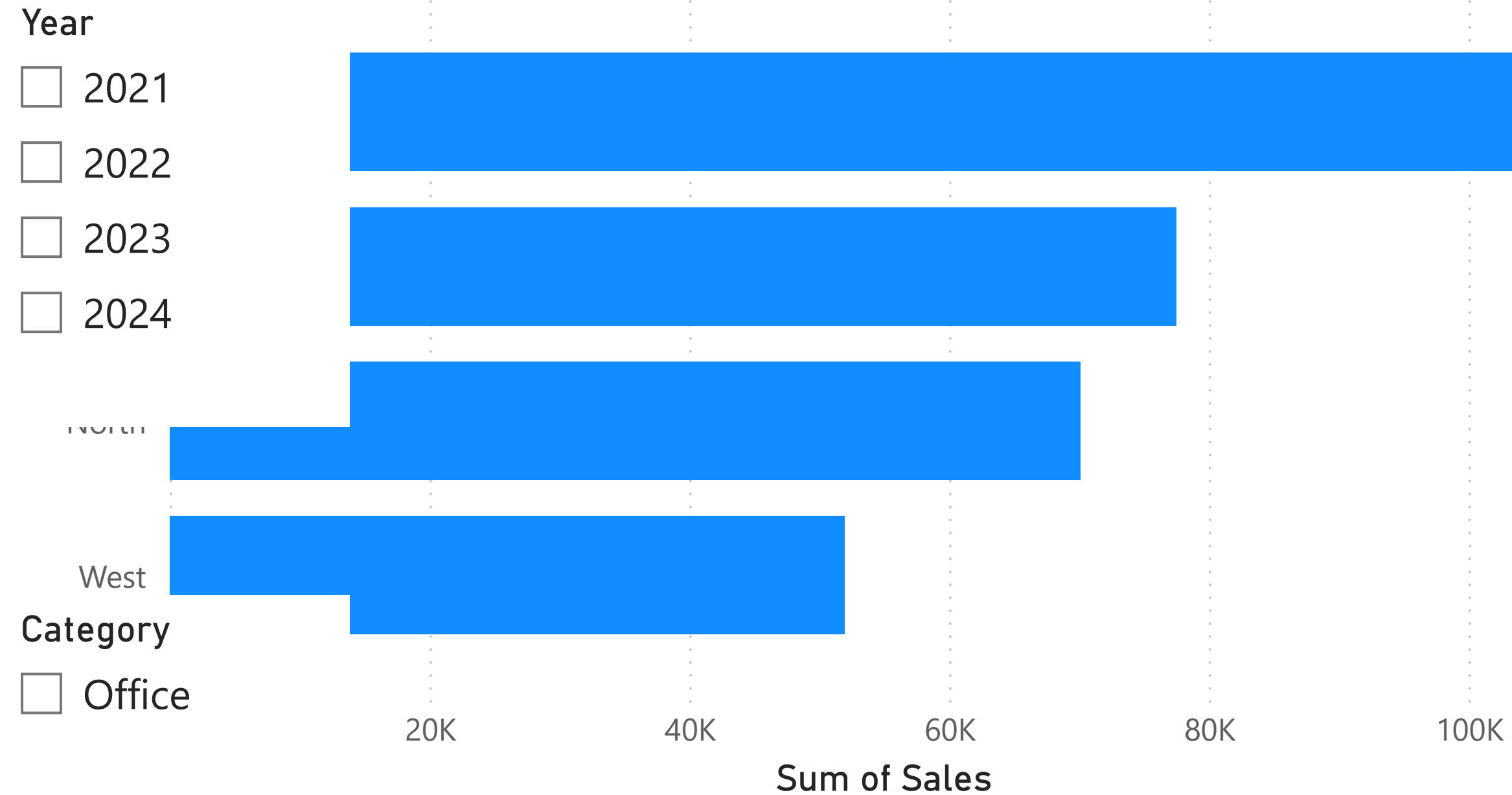
Sum of Sales by Quarter





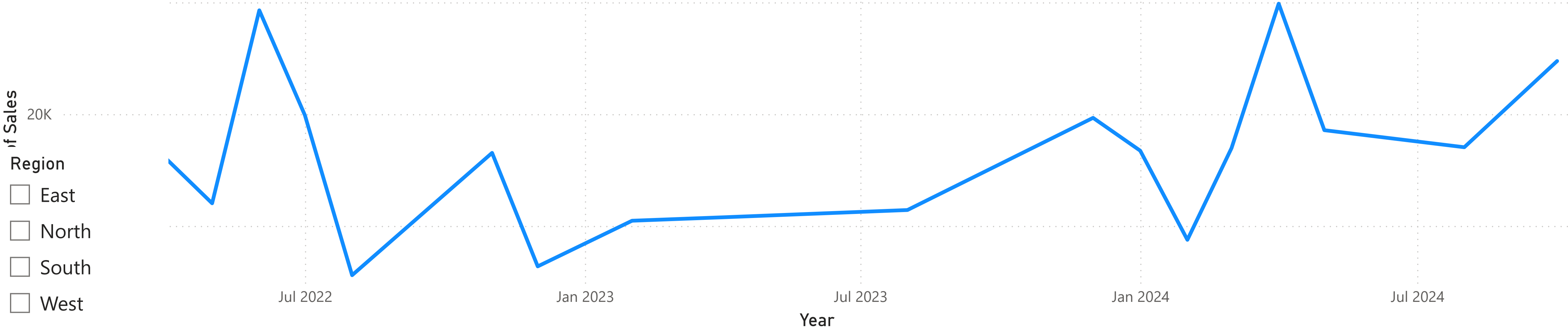
Product Detail

Sum of Sales by Region



Order_ID	Customer_Name	Year	Month	Region	Product	Sum of Quantity	Sum of Sales
1021	Customer_1	2024	May	East	Printer	8	5890
1027	Customer_7	2024	April	East	Printer	5	18412
1047	Customer_27	2022	August	East	Printer	4	2178
1048	Customer_28	2023	August	North	Printer	9	11421
1050	Customer_30	2022	July	South	Printer	9	1806
1065	Customer_45	2022	May	West	Printer	1	9708
1074	Customer_54	2023	February	North	Printer	3	5235
1083	Customer_63	2022	May	South	Printer	6	2316
1085	Customer_65	2024	August	South	Printer	2	17038
1090	Customer_70	2024	April	East	Printer	6	11466
1094	Customer_74	2024	February	West	Printer	4	1376
1097	Customer_77	2024	January	South	Printer	7	8808
1101	Customer_81	2022	June	East	Printer	5	2179
1104	Customer_84	2022	December	South	Printer	6	6381
Total						151	303220

Year, Quarter and Month





Region Detail

626K

Year

☐ 2021

☐ 2022

☐ 2023

☐ 2024

Year, Quarter and Month

72

Count of Order ID

8.69K

Average of Sales

Sum of

20K

Category

☐ Accessories

☐ Electronics

☐ Furniture

☐ Office

Region

Region

West

Region

☐ West

1M

0.2M

0.3M

0.4M

0.5M

0.6M

0.7M

Sum of Sales

Year

Order_ID

Year

Month

Customer_Name

Region

Product

Sum of Quantity

Sum of Sales

1004

2023

April

Devansh

West

Monitor

3

300

1008

2021

August

Harsh

West

Tablet

1

550

1012

2023

July

Lakshay

West

Phone

2

900

1016

2022

November

Prisha

West

Laptop

1

1350

1020

2021

March

Ujjwal

West

Keyboard

5

125

1024

2024

February

Customer_4

West

Keyboard

5

2933

1025

2023

April

Customer_5

West

Laptop

9

6896

1031

2022

July

Customer_11

West

Tablet

2

3652

1033

2023

December

Customer_13

West

Monitor

2

7373

1034

2022

September

Customer_14

West

Monitor

7

12034

1036

2022

August

Customer_16

West

Desk

9

6831

1037

2022

June

Customer_17

West

Laptop

8

2527

Total

315

625797